



Project 1 – Smart Planter

Seunghun Lee

Physical Affordances

- Freestanding body for a stable indoor surface.
- Open top for plant access and maintenance.
- Separate inner pod and water reservoir.
- Rim LED and side buttons for status and control.

User Needs

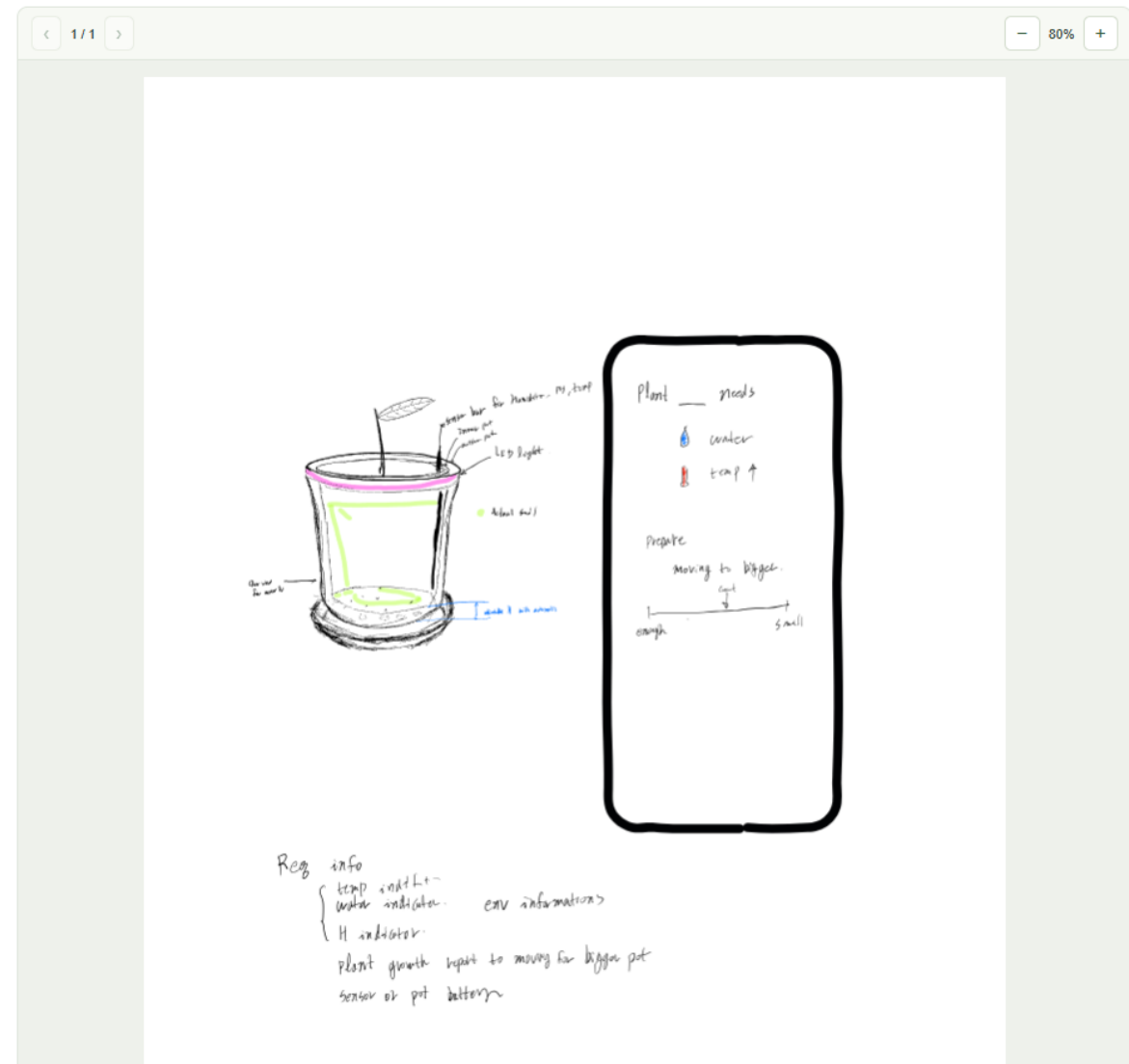
3 Interviews & Findings

	MR. C	MR. A	MR. M
Interview Notes	<i>"I don't always have my phone with me when checking plants. I want a way to care for it right at the planter."</i>	<i>"Checking water levels visually inside the reservoir is tricky. A simple status light would save time."</i>	<i>"When performing maintenance or repotting, automatic movements get in the way and might spill water."</i>
Vanilla Sketch Feedback	Suggested adding physical buttons on the side body for quick access without opening the app.	Appreciated the rim LED status concept; suggested making blinking light patterns for alerts.	Recommended a full pod elevation mode that pauses automatic care during maintenance.

Vanilla Sketches

Design sketch

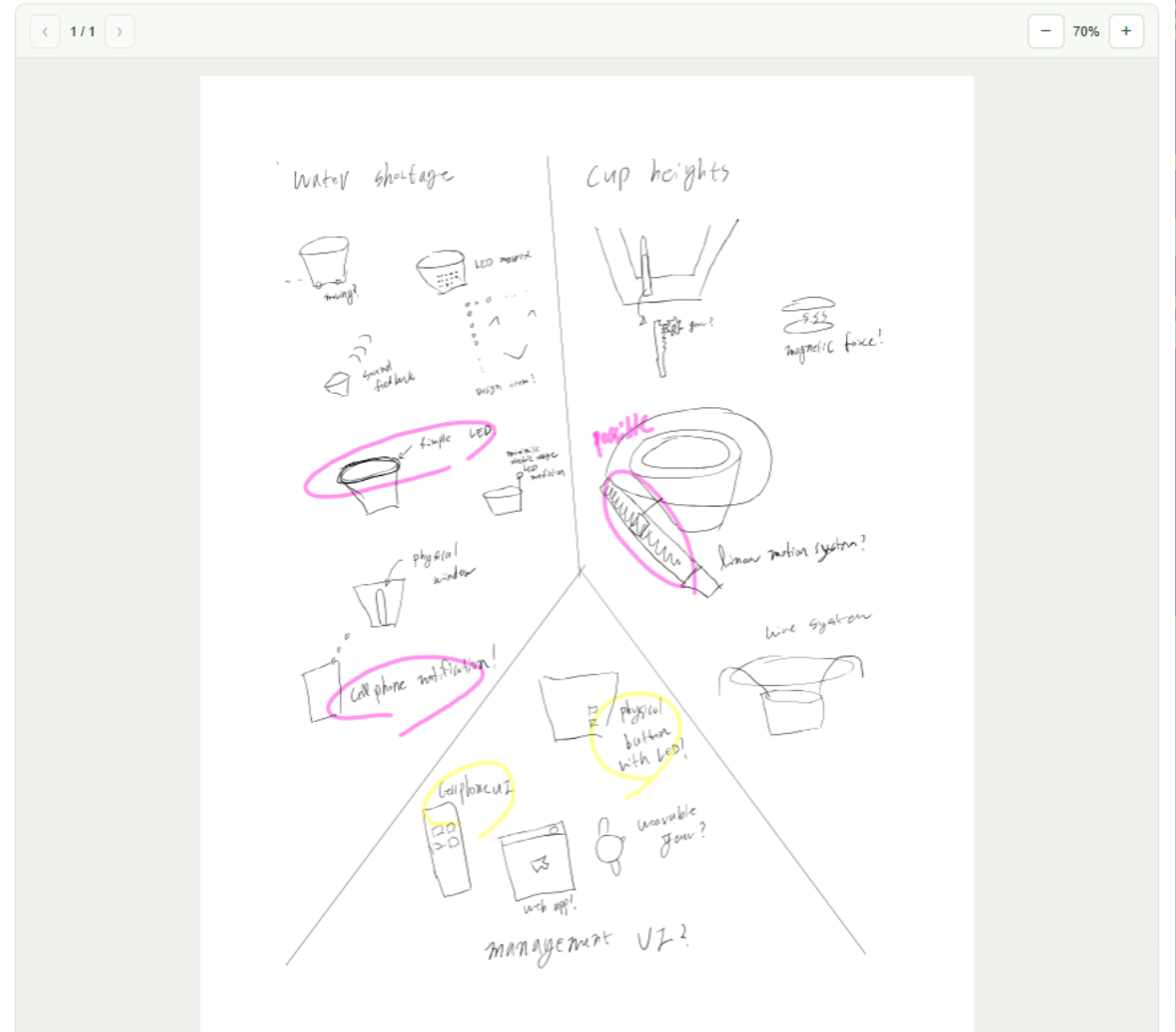
Initial vanilla sketch: the planter structure and companion interface.



10 Minute Sketches

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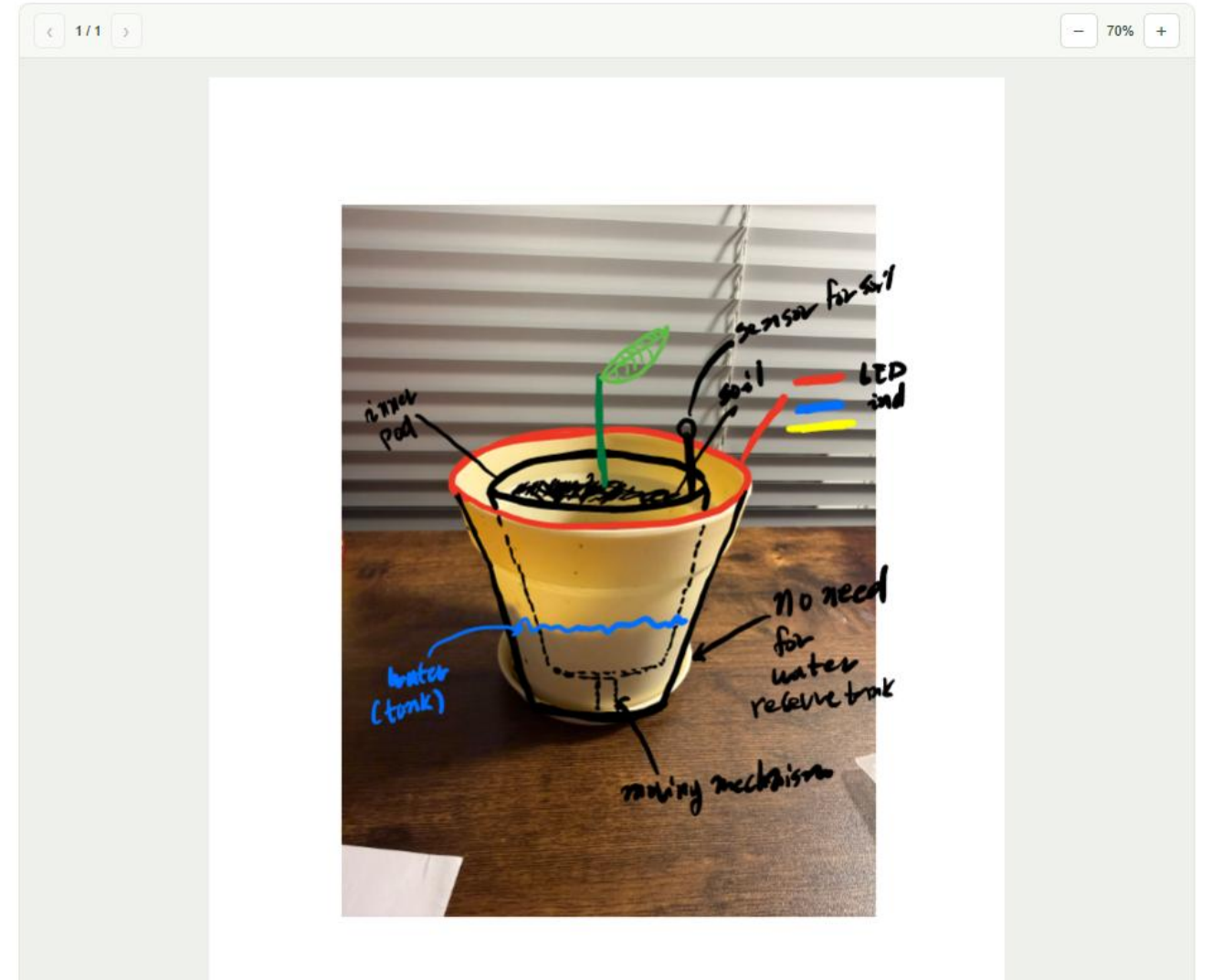
Alternatives for three challenges: feedback about water shortages, cup height adjustment, and the management interface. Highlighted ideas include a simple rim LED, phone notifications, a linear motion mechanism, and physical buttons alongside the phone UI.



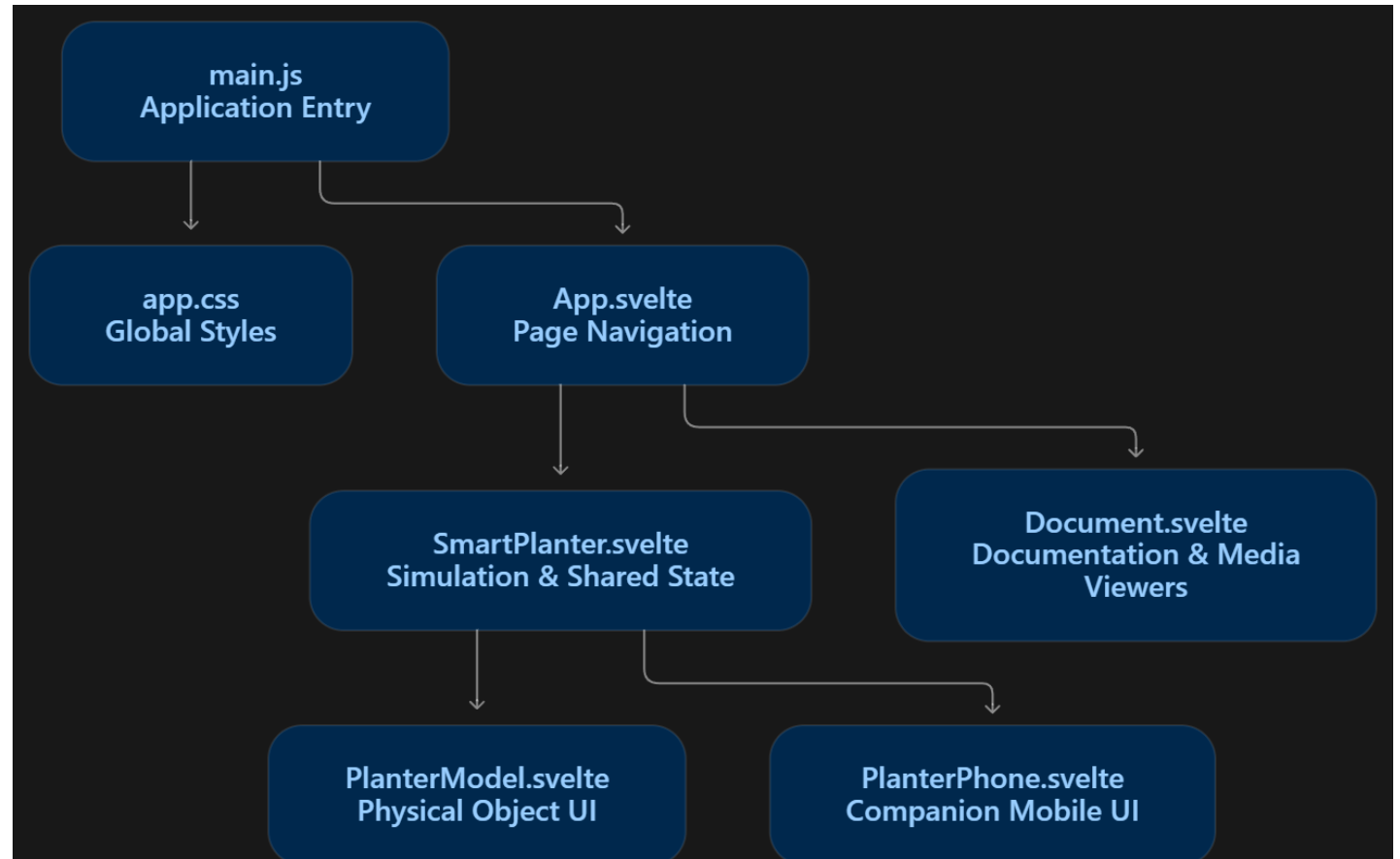
Hybrid Sketches

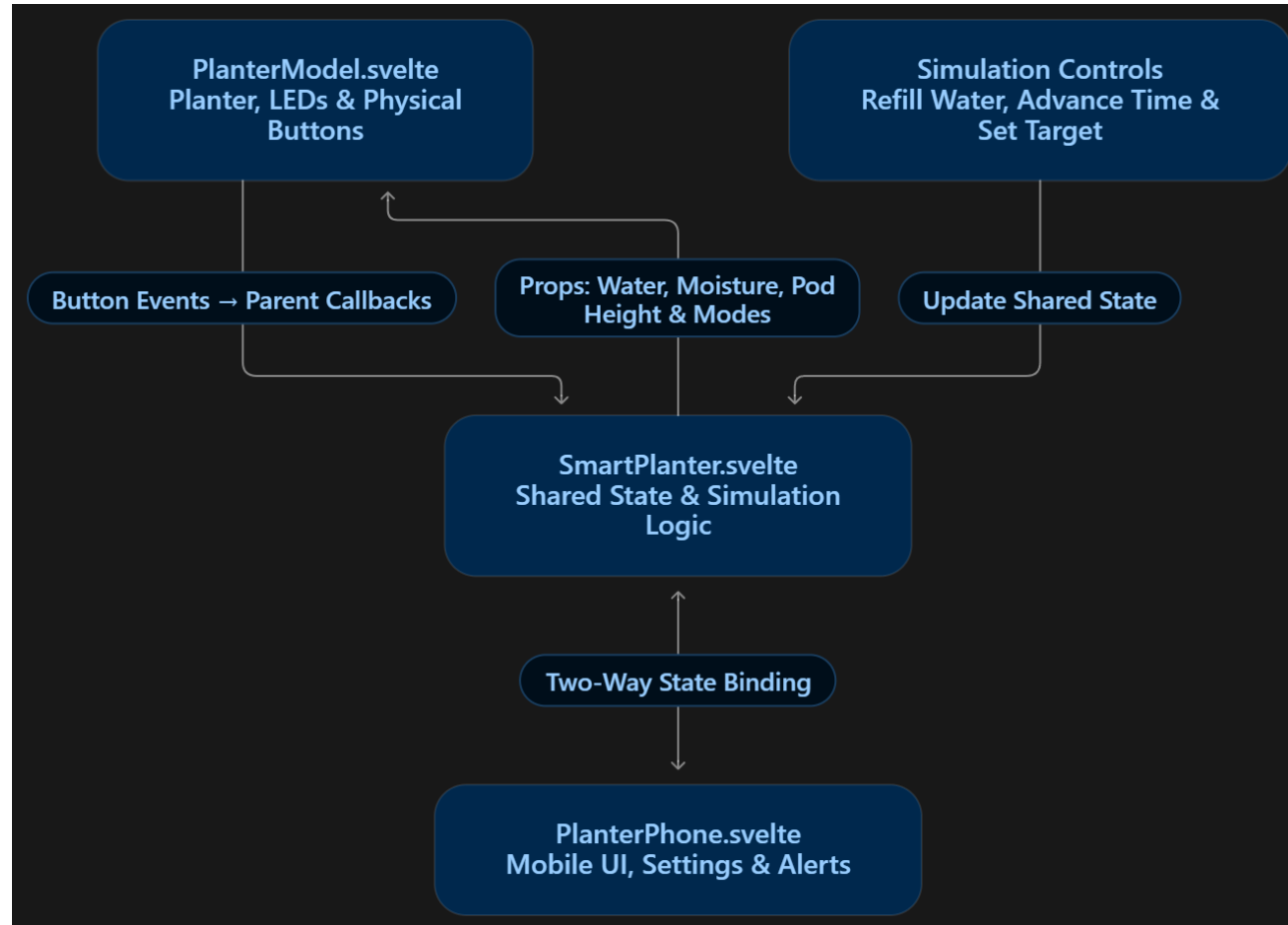
Hybrid sketch

A real planter photograph annotated with a rim LED, soil sensor, inner pod, water reservoir, and movement mechanism. The drawing explores how these elements fit into the physical object. The final prototype also includes side mounted Automatic Care and Maintenance buttons, which are not shown in this sketch.



System Structure





State Flow Diagram

Project 1

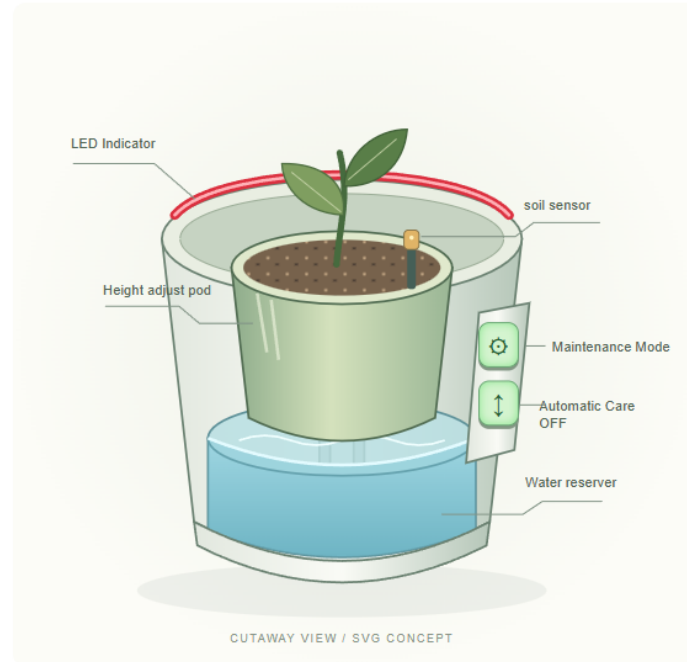
Starter

Document

01 / CONCEPT

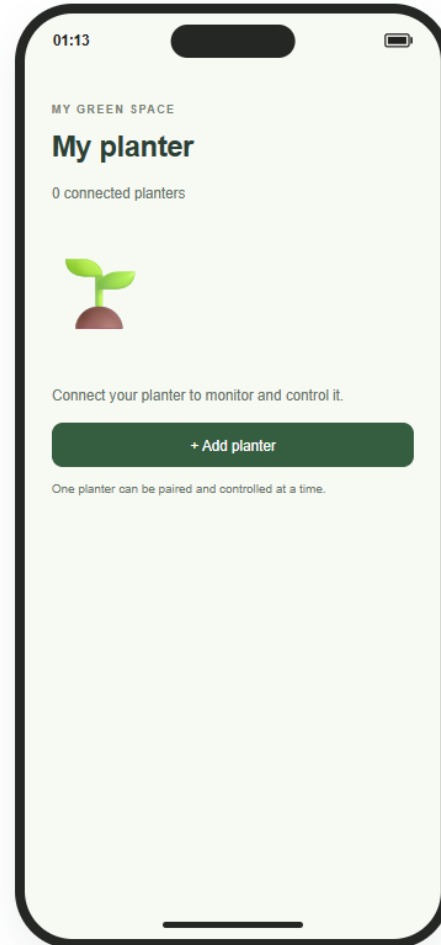
Smart Planter

Cross-section showing the internal structure and height-adjustable cup.



02 / DEVICE UI

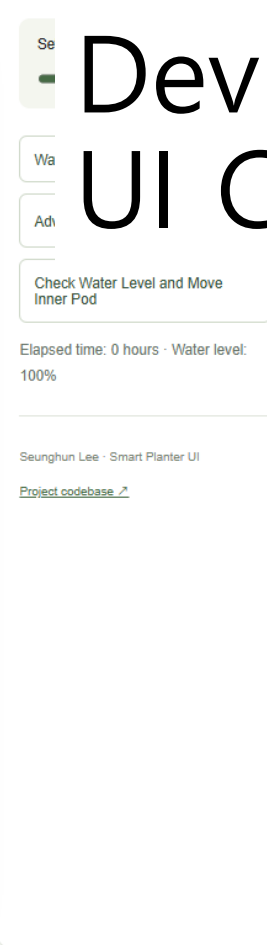
Mobile Screen



03 / CONTROLS

Info

Controls



Device UI & Testing UI Console

<https://ui26.yangdongi.com/project1/>

Future Work

Backend & persistent tracking

Multiple planters & custom plants

Real sensors & device connection

User testing



Questions